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1 CREATE TABLE customers (  
2     customer_id INTEGER PRIMARY KEY,  
3     customer_name TEXT NOT NULL,  
4     city TEXT,  
5     email TEXT  
6 );  
7  
8 CREATE TABLE products (  
9     product_id INTEGER PRIMARY KEY,  
10    product_name TEXT NOT NULL,  
11    category TEXT,  
12    price REAL  
13 );  
14  
15 CREATE TABLE orders (  
16     order_id INTEGER PRIMARY KEY,  
17     customer_id INTEGER,  
18     product_id INTEGER,  
19     quantity INTEGER,  
20     order_date TEXT,  
21     FOREIGN KEY (customer_id) REFERENCES  
    customers(customer_id),  
22     FOREIGN KEY (product_id) REFERENCES  
    products(product_id)  
23 );  
24  
25 INSERT INTO customers VALUES  
26 (1, 'Rahul', 'Bangalore', 'rahul@gmail.com'),  
27 (2, 'Ananya', 'Chennai', 'ananya@gmail.com'),  
28 (3, 'Arjun', 'Mumbai', 'arjun@gmail.com'),  
29 (4, 'Priya', 'Delhi', 'priya@gmail.com'),  
30 (5, 'Sneha', 'Bangalore', 'sneha@gmail.com');  
31
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32 INSERT INTO products VALUES
33 (101, 'Laptop', 'Electronics', 55000),
34 (102, 'Smartphone', 'Electronics', 25000),
35 (103, 'Headphones', 'Electronics', 3000),
36 (104, 'Shoes', 'Fashion', 2000),
37 (105, 'Backpack', 'Fashion', 1500);
38
39 INSERT INTO orders VALUES
40 (1001, 1, 101, 1, '2026-08-01'),
41 (1002, 2, 102, 2, '2026-08-03'),
42 (1003, 3, 103, 3, '2026-08-05'),
43 (1004, 1, 104, 2, '2026-08-07'),
44 (1005, 4, 105, 1, '2026-08-10'),
45 (1006, 5, 102, 1, '2026-08-12'),
46 (1007, 2, 103, 2, '2026-08-15'),
47 (1008, 3, 101, 1, '2026-08-18');
48 SELECT * FROM customers;
49 SELECT product_name, category, price
50 FROM products
51 WHERE price > 3000
52 ORDER BY price DESC;
53 SELECT
54     p.product_name,
55     SUM(o.quantity) AS total_quantity_sold,
56     SUM(o.quantity * p.price) AS total_revenue
57 FROM orders o
58 JOIN products p ON o.product_id = p.product_id
59 GROUP BY p.product_name
60 ORDER BY total_revenue DESC;
61 SELECT
62     AVG(customer_revenue) AS
        average_revenue_per_user
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63 FROM (
64     SELECT
65         o.customer_id,
66         SUM(o.quantity * p.price) AS
        customer_revenue
67     FROM orders o
68     JOIN products p ON o.product_id = p.
        product_id
69     GROUP BY o.customer_id
70 );
71 SELECT product_name, price
72 FROM products
73 WHERE price > (
74     SELECT AVG(price)
75     FROM products
76 )
77 ORDER BY price DESC;
78 SELECT
79     c.customer_name,
80     c.city,
81     o.order_id,
82     o.quantity
83 FROM customers c
84 INNER JOIN orders o
85     ON c.customer_id = o.customer_id
86 ORDER BY c.customer_name;
87 SELECT
88     c.customer_name,
89     c.city,
90     o.order_id,
91     o.quantity
92 FROM customers c
93 LEFT JOIN orders o
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94      ON c.customer_id = o.customer_id
95 ORDER BY c.customer_name;
96 SELECT
97     c.customer_name,
98     c.city,
99     o.order_id,
100    o.quantity
101 FROM orders o
102 RIGHT JOIN customers c
103     ON o.customer_id = c.customer_id
104 ORDER BY c.customer_name;
105 CREATE VIEW customer_orders AS
106 SELECT
107     c.customer_name,
108     c.city,
109     p.product_name,
110     o.quantity,
111     p.price,
112     (o.quantity * p.price) AS revenue
113 FROM customers c
114 JOIN orders o
115     ON c.customer_id = o.customer_id
116 JOIN products p
117     ON o.product_id = p.product_id;
118 SELECT * FROM customer_orders;
119 CREATE INDEX idx_orders_customer
120 ON orders(customer_id);
121 EXPLAIN QUERY PLAN
122 SELECT *
123 FROM orders
124 WHERE customer_id = 2;
```